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PALEOMAGNETISM AND AGE OF MAFIC PLUTONS, WICHITA MOUNTAINS, OKLAHOMA

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Abstract. A paleomagnetic reconnaissance (43 samples) of the Raggedy Mountain Gabbro Group, which is comprised of a layered complex of anorthosites, anorthositic gabbros, and massive gabbro, shows: (1) stable remanent magnetization and (2) consistent directions which differ significantly from those of the nearby Cambrian Wichita Granite Group, with exception of one site near the granite contact. These results suggest a Precambiran age for the gabbro group. The difference in pole positions between the granitic and mafic groups demonstrates that the plutonic episodes responsible for the mafic and granitic magmas are significantly separated in time. Models of formation of the Wichita aulacogen should be altered to account for the difference in time between the mafic and silicic plutonism.

Introduction

The basement rocks under the southern margin of the North American craton yield radiometric dates of Elsonian (1,300 m.y.) to Grenvillian (1,000 m.y.) age (Muehlberger and others, 1966; Bayley and Muehlberger, 1969). The Wichita Mountains of southern Oklahoma (Figure 1) are an apparent exception where the basement, a complicated sequence of igneous rocks, has generally yielded ages between 500 and 535 m.y. (Ham and others, 1964; Burke and others, 1969).

Our concept of the basement relationships is schematically summarized in Figure 2. The Raggedy Mountain Gabbro Group consists of the Glen Mountains Layered Complex (Powell and others, 1980), a sequence of troctolites, anorthosites and anorthositic gabbros (Hunter, 1973; Powell and Phelps, 1977) and the Roosevelt Gabbros (Powell and others, 1980), comprised of cross-cutting massive gabbro bodies. This group is nonconformably overlain by the Navajo Mountain Basalt-Spilite Group and Carlton Rhyolite Group, but all are intruded by a series of granitic plutons, collectively known as the Wichita Granite Group. In addition, there are minor intrusions of intermediate bulk composition (Huang, 1955) and uncertain relationships, such as the Cold Springs Breccia (Powell and others, 1980).

The Carlton Rhyolite Group and Wichita Granite Group are Cambrian in age, established by Pb-Pb and U-Pb dates on zircons (Tilton and others, 1962) and by K-Ar dating (Ham and others, 1964). The results of paleomagnetic work and pole positions (e.g. Ku and Scharon, 1966; Ku and others, 1967; Spall, 1968, 1970; Vincenz and others, 1975) are consistent with this age.

Although cross-cutting relationships show the Raggedy Mountain Gabbro Group to be older than the Wichita Granite Group, the absolute age of the mafic rocks (Figure 2) is controversial. Three out of four analyses yielded Cambrian K-Ar and Rb-Sr ages from 509 to 535 m.y. (Ham and others, 1964; Burke and others, 1969), but the fourth yielded a late Proterozoic age (730 m.y., Burke and others, 1969). Accordingly, Ham and others considered the mafic plutonism to be an earlier phase of a single Cambrian magmatic event. Their view has been widely accepted (e.g., Hoffman and others, 1974; Bayley and Muehlberger, 1968), and this magmatism was interpreted as an early event in the formation of the southern Oklahoma aulacogen.

Powell and Fischer (1976) and Powell and Phelps (1977), however, favor a Precambrian age for the mafic plutonism for two reasons: (1) petrologic studies show that these rocks cannot be consanguineous with the overlying granitic suite; and (2) the unconformity between the suites is much greater than inferred by Ham and others. Powell and Phelps (in contradiction to Scofield, 1973; 1975) hold that the layered rocks of the Wichita Mountains correspond to the middle part of such classical complexes as the Stillwater and Bushveld and that a greater time span than allowed by the radiometric age dates is necessary to account for the removal of the overlying strata. Thus, the geological and petrological relations suggest that the Raggedy Mountain Group is of Precambrian age and that the radiometric ages (K-Ar and Rb-Sr) are not primary but the result of a Cambrian thermal overprint.

Paleomagnetism offers an independent approach to these problems. Previous magnetic work on the mafic rocks showed that their directions differed materially from those derived from the granites but little significance was attached to this date (Ku and others, 1967; Vincenz and others, 1975). We here report on a magnetic reconnaissance of some of the mafic rocks of the Wichita Mountains. The work was carried out with attention to their geology, petrographic differentiation, structural attitude, and proximity to intrusive contacts.

Our data may be grouped into four lots (Table 1; Figures 1 and 3); these include: (1) the main part of the Glen Mountains Layered Complex regionally dipping 15 to 20° to the NE as defined by the igneous lamination (localities E 2, 3; W 4, 5, 6, 7); (2) massive Mount Sheridan Gabbro, one of the Roosevelt Gabbros (Powell and others, 1980) at the foot of Mt. Sheridan (E-3); (3) aerially limited portions of the Glen Mountains Layered Complex at Panther Creek (E-1) whose attitude (dipping 40° to the SW) differs substantially from that of the regional orientation of the group and are near the contact with the Wichita Granites; (4) diabase dikes intrusive into the Glen Mountains Layered Complex (W-5). With exception of the diabase samples (magnetically unstable and not considered further), these rocks were found to have high magnetic stability, and to yield geologically consistent results. While the number of localities examined is small, the results provide conclusions and encourage further work.

Methods

Samples, oriented by sun compass, were progressively AF demagnetized in peak fields as high as 700 oe and measured on a PAR spinner magnetometer. The vector chosen for any one sample was the one corresponding to the demagnetization step showing the smallest ratio of vector shift to change in field intensity (similar to the stability criterion of Symons and Stupavsky, 1974). Thermal demagnetization was performed on 12 samples which had been AF-demagnetized to 300 oe. Structural corrections were made on the assumption that the layering in the rocks was initially horizontal, and that primary magnetization preceded tilting. For most of the localities the correction is small (about 15° NE), but at Panther Creek (E-1) it is large, and as will be shown, the magnetization here seems to either postdate tilting or the assumption of initial horizontality is not warranted.

Nature and Stability of Magnetization

The bulk of the samples studied are troctolites, anorthosites and anorthositic gabbros from the layered complex and show

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Table 1. Paleomagnetic Directions

Sample Set	Sites	Samples	Polarity	Average Median Destructive Field	Structural Corrections	NRM Dec	NRM Inc	NRM α95	AF Demagnetized Mean Dec	AF Demagnetized Mean Inc	R	k	α95
W Wichitas Layered C	4 W 4-7	20	mixed	322 oe	uncoor / corr	261 9 / 262 0	−5 6 / +6 8	90 0	232 5 / 236 4	+15 8 / +34 1	3 932	44 4	13 9
E Wichitas Layered C	4 E 2-5	10	mixed	272 oe	uncorr / corr	237 5 / 242 1	+15 6 / +20 8	49 4	232 5 / 233 2	−0 5 / +6 3	3 867	22 5	19 8
Mean of Struct Normal layered rocks	8	33		297 oe	uncorr / corr	246 3 / 250 5	+11 8 / +20 1	47 3 / 45 5	234 6 / 236 8	+7 3 / +20 7	31 081 / 30 463	16 7 / 12 6	6 3 / 7 3
Mt Sheridan Gabbro	1 E-3	3	normal	253 oe	uncorr / corr	162 9 / 156 8	+58 9 / +73 7	90 0	239 6 / 238 6	−7 6 / −2 2	2 972	72 4	14 6
Panther Creek Layered C	1 E-1	6	reversed	230 oe	uncorr / corr	102 7 / 127 6	+23 8 / +47 9	20 6	80 0 / 104 3	+29 9 / +64 9	5 577	11 8	20 3

excellent magnetic stability. Alternating field demagnetization removed a viscous component in peak fields less than 300 oe, and further AF treatment in fields up to 700 oe caused little directional movement. Subsequent thermal demagnetization produced little change in intensity or directions up to the 525-585°C range whereupon nearly all of the remanence was lost. This behavior, similar to that of samples from the Stillwater Complex (Bergh, 1970), suggests that the remanence resides primarily in magnetite.

Magnetite is sporadically present as discrete grains rarely up to 10 cm across in some units of the Glen Mountains Layered Complex, but also occurs as inclusions in plagioclase. Numerous mineral inclusions have exsolved from the feldspar, forming rod-shaped grains elongated along the (001) axis of the plagioclase. The saturation magnetization of a plagioclase separate shows the presence of a mineral which saturates magnetically in a field below 2 kOe. The J_s/T behavior of the plagioclase concentrate shows a well-defined Curie point at approximately 580°C. Upon cooling the data are nearly repeatable. The Curie point strongly indicates that the dominant magnetic phase within the plagioclase is almost pure magnetite.

The magnetic behavior of the whole rock and plagioclase concentrates demonstrates: a) saturation in low fields, b) a Curie point of 580°C, and c) blocking temperatures below 600°C and leads us to conclude that the stable magnetization resides mainly in fine magnetite inclusions dispersed through the feldspar although orthopyroxene from these samples also have inclusions and may also contribute to the remanence. This mode of magnetization of the feldspars is similar to that of the Lambertville sill described in detail by Hargraves and Young (1969). The experimental thermal behavior of layered complex rocks implies long term stabilities at temperatures of up to 450 or 500° (cf. Pullaiah and others, 1975). Hence, a thermal remagnetization of these rocks is only likely in proximity to contacts with large intrusives such as the Wichita Granites.

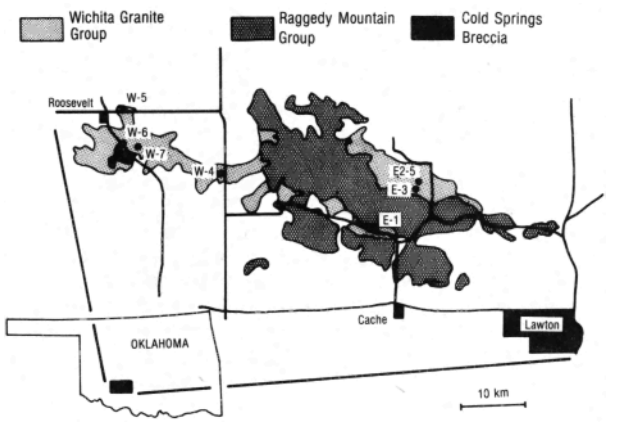


Fig. 1. Sketch map of Wichita Mountains, showing sample locations. E2-5: A series of closely clustered sites in the layered complex, grouped as "Eastern Wichitas." W4-7: a series of sites in the layered complex, grouped as "Eastern Wichitas." E-1: layered complex with anomalous dip, at Panther Creek. E-3: Mount Sheridan Gabbro. Unstable diabase from loc. W-4. Base map after Powell and others (1980).

Remanence Directions

The magnetic vectors obtained are shown in Table 1 and plotted in Figure 3. The petrologic-magnetic units are:

(1) Regionally dipping Glen Mountains Layered Complex. In the eastern and western parts of the Wichita Mountains, the layered complex generally dips northward at from 15° to 20°. The 20 samples from the western Wichitas show mixed polarities (Figure 3a), whereas, the ten samples from eastern part of the range show normal ones with only one exception.

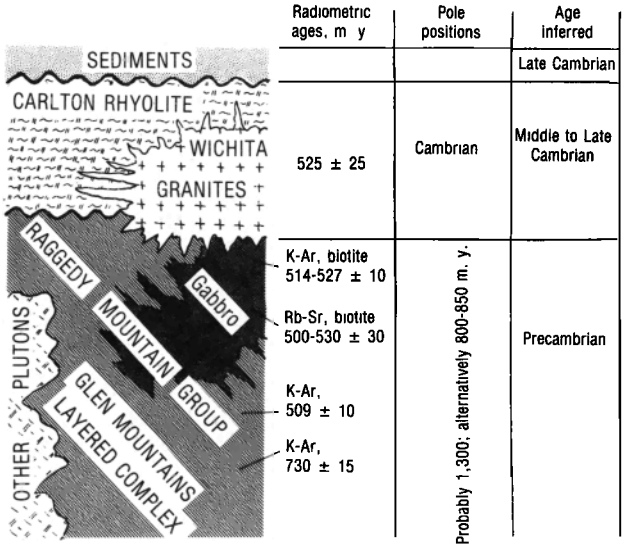


Fig. 2. Schematic concept of age relations in the basement of the Wichita Mountains. Unit labelled as "Gabbro" refers to the inclusive term Roosevelt Gabbro which includes the Mount Sheridan, Sandy Creek, and Glen Creek Gabbros. Terminology and ages are from the summary by Powell and others (1980, p. 1887).

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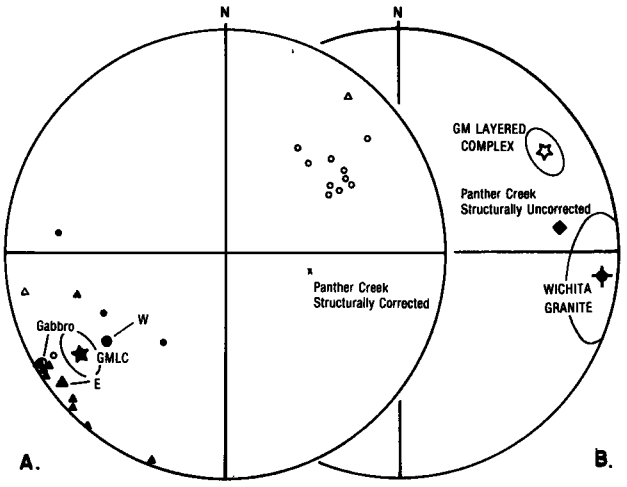


Fig. 3 Remanence directions obtained from Raggedy Mountain Group, Wichita Mountains, equal area projection, open symbols on upper himisphere.

A. Four sets of samples corrected for structual tilt. For computation of mean directions, northeast-pointing directions have been reversed: Triangles: Individual samples and mean from layered complex of sites (E) in eastern Wichita Mountains. Circles: Individual samples and mean for sites (W) in western Wichita Mountains. Combined mean shown by star (GMLC). Gabbro: Mean of three samples of massive Mount Sheridan Gabbro. Panther Creek: Mean from anomalous site at Panther Creek.

b. Mean for the anomalously dipping at Panther Creek (E-1) near the granite contact, uncorrected for tilt. For comparison, mean for other sites (reverse pole shown), and mean for Wichita granites (from Spall) are shown with their 95 per cent cones of confidence.

(2) Mount Sheridan Gabbro. Three samples of massive gabbro were taken at a locality near the foot of Mount Sheridan. These yield a vector mean very close to that of the layered complex in the eastern Wichita Mountains (Figure 3a). This suggests to us that the Mount Sheridan Gabbro was closely associated in origin with the layered complex. Whether this is true for other gabbroic bodies in the Wichita Mountains remains in question.

(3) Glen Mountains Layered Complex at Panther Creek. The Panther Creek locality differs in that the layering dips southwestward at about 40° and seems to have been close to the contact with the granite, which crops out nearby. The magnetic vectors of these rocks are anomalous. If corrected for dip (Figure 3a), the vector mean shows an improbably high incination, suggesting that the magnetization was at least in part imposed after structural tilting. The vector uncorrected for structural tilt (Figure 36)shows reasonable inclination, but remains anomalous in declination, pointing downward almost due east. This pole lies near the 95 per cent cone of confidence of Spall's pole for the (Cambrian) granites (Figure 3b). The data from this locality suggests that these layered rocks were either remagnetized by the nearby granitic intrusives, in Cambrian time or have undergone a complex series of tectonic rotations prior to initial acquisition of magnetization. Alternatively, the initial assumption of horizontality of the layering may not apply. McBirney and Noyes (1979) demonstrated that layering in plutons such as these often develops parallel to cooling surfaces and not necessarily as a result of gravity settling, but differential structural rotations would still be necessary to bring the Panther Creek locality into agreement with the main portion of the layered complex.

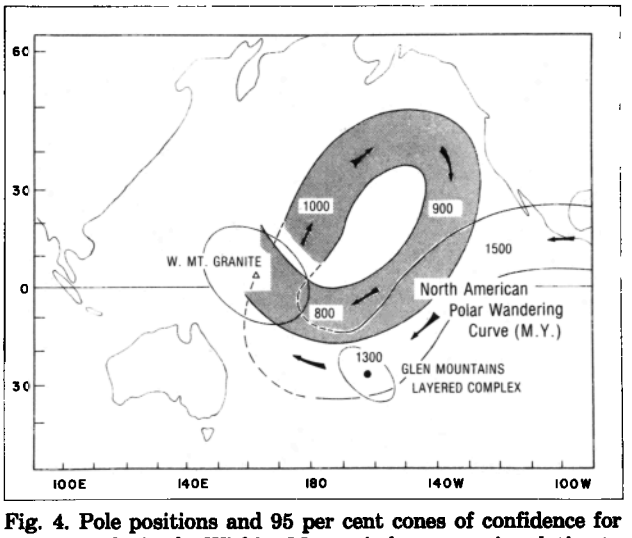


Fig. 4. Pole positions and 95 per cent cones of confidence for various rocks in the Wichita Mountain basement, in relation to the polar wander path of Irving and McGlynn (1976). Open triangle is the Cambrian Wichita Granite pole of Spall with its associated cone of confidence.

Discussion

Panther Creek is not the only locality near granitic rocks. Locality W-6 is less than 10 meters above the contact with the Cold Springs Breccia although the lack of thermal overprint in this case may be due to the small volume of the Cold Springs Breccia involved.

The poles to the units investigated plotted in relation to the polar wandering curve of Irving and McGlynn (1976) (Figure 4) suggest 1,300 m.y. as the most likely age for the Raggedy Mountain Group. The looped path of the polar wander curve and the diameter of the cone of 95 per cent confidence, however, leaves room for the possibility of an age near 800 m.y. (coinciding with a K-Ar age by Burke and others, 1969, for the layered complex, see Figure 2). In any case, the Raggedy Mountain poles are significantly different from those derived from the Wichita Granites.

Conclusions

(1) The remancence directions of the Raggedy Mountain Group are significantly different from those of the Wichita Granites (Spall, 1968; 1970), indicating that they are of different age.

(2) Distant from contacts with known Cambrian granites the Raggedy Mountain rocks yield vectors of mixed polarity, pointing to a pole in the 1,300 m.y. sector of the Irving-McGlynn polar wander curve. Because of the subsequent loop in this curve, the possibility of an age of around 800 m.y. cannot be excluded, but in any case the Raggedy Mountain Group was emplaced in Precambrian time.

(3) A locality near the contact with Cambrian granites was probably remagnetized in Cambrian time, at temperatures in excess of 450°. We suggest that argon and strontium retention in the basic rocks was relaxed over a somewhat broader zone, and that the radiometric dates reported from them record this Cambrian heating rather than their true age.

(4) These data do not invalidate the aulacogen concept (Shatsky, 1946) but call for a longer and more episodic evolution than hitherto believed.

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